TO'LIQ ENSIKLOPEDIK MA'LUMOTNOMA VA QO'LLANMA

MATHPROFI.RU VA OLIY MATEMATIKA: Barcha Maxsus Mavzular, Oltin Formulalar, Jadvallar va Yechish Sxemalari

mathprofi.ru portalining barcha 14 ta bo'limi (245+ ma'ruza), 63 ta maktab kursi darsi va 12 ta maxsus PDF ma'lumotnomasi asosida tayyorlangan to'liq o'zbekcha konspekt-qo'llanma. Barcha formulalar, qiyosiy jadvallar, algoritmlar va qadamma-qadam yechilgan universitet namunalari jamlangan.

13 ta	37 ta	12 ta	350+
Fundamental bob	Chuqurlashtirilgan mavzu	Oltin shpargalka jadvali	Formulalar va misollar

Tuzuvchi: Oliy Matematika AI & Aleksandr Yemelin (mathprofi.ru) metodikasi asosida

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I BOB

Mathprofi.ru va Oliy Matematika Portalining To'liq Qiyosiy Tahlili

Ushbu fundamental qo'llanma **mathprofi.ru** (muallif Aleksandr Yemelin) portalidagi barcha **14 ta oliy matematika bo'limi (245+ ma'ruza)**, **63 ta maktab kursi («Песочница»)** hamda **12 ta maxsus PDF ma'lumotnoma («Математические формулы и таблицы»)** ni chuqur tahlil qilish asosida tuzilgan. Unda bizning asosiy platformamizdagi 136 ta darsni to'ldiruvchi **39 ta maxsus mavzu, barcha oltin formulalar jadvallari, tipovoy masalalarni yechish algoritmlari va namunaviy misollar** yagona ensiklopedik kitob holiga keltirilgan.

📊 **Qiyosiy qamrov ko'rsatkichlari**

- **Mathprofi.ru umumiy bazasi:** 14 ta oliy matematika bo'limi + Maktab kursi + 12 ta PDF shpargalka jadvali.
- **Saytimizda (oliy-matematika.vercel.app) mavjud:** 14 ta modul doirasida 136 ta interaktiv dars, 189 ta formula kartochkasi, AI misol yechuvchi, Matritsa va Vektor kalkulyatorlari.
- **Ushbu PDF kitobga kiritilgan:** Saytda alohida sahifa sifatida ajratilmagan yoki Mathprofi.ru da maxsus PDF-jadvallar / tor ixtisoslashgan ma'ruzalar ko'rinishida berilgan **barcha 39 ta chuqurlashtirilgan mavzu va 12 ta to'liq ma'lumotnoma jadvali**!

1.1. Bo'limlar bo'yicha to'liq qiyosiy xarita

№	Mathprofi.ru bo'limi	Ushbu PDF qo'llanmada to'liq yoritilgan maxsus mavzular va jadvallar
1	**Повторяем школьный курс & Горячие формулы**	Ko'p qavatli kasrlar qoidalari, Qisqa ko'paytirish (n-daraja), Daraja/Ildiz/Logarifmning 22 xossasi, $0^\circ - 360^\circ$ trigonometrik jadval, Keltirish formulalari, Arkfunksiyalar jadvali.
2	**Аналитическая геометрия**	Yangi bazisga o'tish va koordinata o'qlarini burish (α), Tekislikda chiziqli tengsizliklar sohasi, Tekislikda uchburchakning 12 elementini topish tipovoy sxemasi, 2-tartibli chiziqni burish va invariantlar (I_1, I_2, I_3) bilan kanonik shaklga keltirish.
3	**Высшая алгебра**	ChTS ni turli bazislarda yechish (tayanch yechimlar va Jordan-Gauss jadvali), Kvadratik formani ortogonal almashtirish ($X = QY$) bilan kanonik shaklga keltirish, Kompleks koeffitsiyentli tenglamalar va ChTS.
4	**Пределы**	Barcha 5 ta ajoyib limit va 14 ta ekvivalentlik jadvali, $t = x - a$ almashtirish va keltirish formulalari bilan yuqori qiyinlikdagi limitlar, Makloren qatorlari yordamida limit hisoblash.
5	**Производная и Функции**	Oddiy va murakkab funksiyalar hosilalarining 22 qatorli jadvali, Funksiyani to'liq tekshirish va grafik yasashning 8 bosqichli universal algoritmi.
6	**Функции нескольких переменных**	Ikki karrali (umumiy) limit va takroriy limitlar, Yo'nalishlar bo'yicha limit mavjud emasligi, Uzluksizlik va uzilish chiziqlari, Xususiy hosilali differensial tenglamalarni tekshirish, Eng kichik kvadratlar usuli (EKU).
7	**Однократные интегралы**	26 bandli integrallar jadvali va differensial ostiga kiritish, Kvadrat uchhadli kasrlar va Ostrogradskiy usuli, Trigonometrik integrallar xaritasi, Chebishevning binomial differensiallari, Aniq/xosmas integrallarning effektiv usullari.
8	**Дифференциальные уравнения**	Tartibini pasaytirish mumkin bo'lgan 3 turdagi DT, Yuqori tartibli ($n \geq 3$) chiziqli DT, O'ng tomon bo'yicha xususiy yechim tanlashning 12 qatorli Mathprofi jadvali, Geometrik va fizik masalalardan DT tuzish.
9	**Числовые и функциональные ряды**	Funksional qatorlarning tekis yaqinlashishi va Veyershtrass alomati, Makloren qatorlariga yoyish va binomial qatorning barcha xususiy hollari jadvali.
10	**Кратные и поверхностные интегралы**	Tekis shakl va jismning massasi, statik momentlari, og'irlik markazi (x_C, y_C, z_C) va inersiya momentlari, 1- va 2-tur sirt integrallari.
11	**Комплексный анализ (ТФКП)**	Kompleks tekislikda chiziqlar va bir/ko'p bog'lamli sohalar, Kompleks funksiya limiti va hosilasi, Konform akslantirishlar ($k = \lvert f'(z_0)\rvert$, $\alpha = \arg f'(z_0)$), Laplas originallar va tasvirlarning 24 qatorli jadvali.
12	**Теория вероятностей и Мат. статистика**	Takrorlanuvchi va takrormas kombinatorika formulalari, Barcha taqsimotlar jadvali, Statistik ehtimol, Asimmetriya (A_s) va Ekstsess (E_k), Takroriy va takrormas tanlanma baholari, Guruhlash turlari, Fexner koeffitsiyenti (K_Φ), Ko'p omilli chiziqli regressiya, Dinamika qatorlari ko'rsatkichlari.

II BOB

Maktab Kursi «Qaynoq Formulalari» va Trigonometrik Jadvallar

Manba: `goryachie_formuly.pdf`, `trigonometricheskie_formuly.pdf`, `trigonometricheskie_tablicy.pdf` va `pesochnica/index.html`. Oliy matematikada limitlar, hosilalar, integrallar va differensial tenglamalarni yechishda har qadamda kerak bo'ladigan fundamental baza.

2.1. Ko'p qavatli (uch qavatli) kasrlarni soddalashtirish qoidalari

Limitlar, hosilalar va integrallarda ko'p qavatli kasrlar hosil bo'lganda asosiy kasr chizig'iga nisbatan quyidagi 4 ta oltin qoida qo'llaniladi:

$$\frac{\frac{a}{b}}{c} = \frac{a}{b \cdot c}, \qquad \frac{a}{\frac{b}{c}} = \frac{a \cdot c}{b}, \qquad \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a \cdot d}{b \cdot c}, \qquad a \pm \frac{b}{c} = \frac{a \cdot c \pm b}{c}$$

2.2. Qisqa ko'paytirish formulalari va Kvadrat uchhad

Formula nomi	Algebraik ifodasi	Oliy matematikada qo'llanilishi
Yig'indi va ayirma kvadrati	$(a \pm b)^2 = a^2 \pm 2ab + b^2$	Integralda to'la kvadrat ajratish, 2-tartibli chiziqlar
Kvadratlar ayirmasi	$a^2 - b^2 = (a - b)(a + b)$	Irratsional limitlarda qo'shmaga ko'paytirish ($\sqrt{u} - \sqrt{v}$)
Yig'indi va ayirma kubi	$(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$	Hosilalar, kubik qavslarni ochish
Kublar yig'indisi va ayirmasi	$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$	$\sqrt[3]{u} \pm \sqrt[3]{v}$ limitlarda to'liqsiz kvadratga ko'paytirish
n-darajalar ayirmasi	$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$	Ixtiyoriy n-ildizli limitlarni qisqartirish
Kvadrat uchhadni ajratish	$ax^2 + bx + c = a(x - x_1)(x - x_2), \quad D = b^2 - 4ac$	$[0/0]$ limitlar va ratsional kasrlarni integrallash
To'la kvadratni ajratish	$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a}$	Kasr va ildizli integrallarni jadval integrallariga keltirish

2.3. Daraja, Ildiz va Logarifmlarning 22 ta asosiy xossasi

Daraja va Ildiz xossalari ($a, b > 0$)

- $a^0 = 1 \ (a \neq 0), \quad a^{-n} = \dfrac{1}{a^n}$
- $a^m \cdot a^n = a^{m+n}, \quad \dfrac{a^m}{a^n} = a^{m-n}$
- $(a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n, \quad \left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$
- $\sqrt[n]{a^m} = a^{\frac{m}{n}}, \quad \sqrt[n]{\dfrac{1}{a^m}} = a^{-\frac{m}{n}}$
- $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}, \quad \sqrt[n]{\sqrt[k]{a}} = \sqrt[nk]{a}$
- **Juft ildiz qoidasi:** $\sqrt{x^2} = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

Logarifm xossalari ($a > 0, a \neq 1, x, y > 0$)

- $a^{\log_a x} = x, \quad \log_a 1 = 0, \quad \log_a a = 1$
- $\ln x = \log_e x \ (e \approx 2.71828), \quad \lg x = \log_{10} x$
- $\log_a(xy) = \log_a x + \log_a y$
- $\log_a\left(\dfrac{x}{y}\right) = \log_a x - \log_a y$
- $\log_a(x^p) = p \log_a x, \quad \log_{a^q}(x^p) = \dfrac{p}{q} \log_a x$
- **Yangi asosga o'tish:** $\log_a x = \dfrac{\ln x}{\ln a}, \quad \log_a b = \dfrac{1}{\log_b a}$

2.4. Eng ko'p qo'llaniladigan Trigonometrik Formulalar (`trigonometricheskie_formuly.pdf`)

1. Asosiy ayniyatlar

$$\sin^2\alpha+\cos^2\alpha=1$$

$$\tan\alpha=\frac{\sin\alpha}{\cos\alpha},\quad \cot\alpha=\frac{\cos\alpha}{\sin\alpha},\quad \tan\alpha\cdot\cot\alpha=1$$

$$1+\tan^2\alpha=\frac{1}{\cos^2\alpha},\qquad 1+\cot^2\alpha=\frac{1}{\sin^2\alpha}$$

2. Qo'shish formulalari

$$\sin(\alpha\pm\beta)=\sin\alpha\cos\beta\pm\cos\alpha\sin\beta$$

$$\cos(\alpha\pm\beta)=\cos\alpha\cos\beta\mp\sin\alpha\sin\beta$$

$$\tan(\alpha\pm\beta)=\frac{\tan\alpha\pm\tan\beta}{1\mp\tan\alpha\tan\beta}$$

3. Ikkilangan burchak va Darajani pasaytirish

$$\sin 2\alpha=2\sin\alpha\cos\alpha$$

$$\cos 2\alpha=\cos^2\alpha-\sin^2\alpha=2\cos^2\alpha-1=1-2\sin^2\alpha$$

$$\sin^2\alpha=\frac{1-\cos 2\alpha}{2},\qquad \cos^2\alpha=\frac{1+\cos 2\alpha}{2}$$

$$1-\cos\alpha=2\sin^2\frac{\alpha}{2},\qquad 1+\cos\alpha=2\cos^2\frac{\alpha}{2}$$

4. Ko'paytmani yig'indiga va Yig'indini ko'paytmaga

$$\sin\alpha\cos\beta=\frac{1}{2}[\sin(\alpha-\beta)+\sin(\alpha+\beta)]$$

$$\cos\alpha\cos\beta=\frac{1}{2}[\cos(\alpha-\beta)+\cos(\alpha+\beta)]$$

$$\sin\alpha\sin\beta=\frac{1}{2}[\cos(\alpha-\beta)-\cos(\alpha+\beta)]$$

$$\sin\alpha\pm\sin\beta=2\sin\frac{\alpha\pm\beta}{2}\cos\frac{\alpha\mp\beta}{2}$$

$$\cos\alpha+\cos\beta=2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$

$$\cos\alpha-\cos\beta=-2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$$

2.5. Trigonometrik va Teskari Trigonometrik Jadvallar (`trigonometricheskie_tablicy.pdf`)

Gradus va Radian o'rtasida o'tish formulalari: $\alpha_{\text{rad}}=\frac{\pi}{180^\circ}\cdot\alpha_{\text{grad}},\qquad \alpha_{\text{grad}}=\frac{180^\circ}{\pi}\cdot\alpha_{\text{rad}}.$

Burchak α (grad)	0°	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
α **(radian)**	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π
$\sin\alpha$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0
$\cos\alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1	0	1
$\tan\alpha$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	—	$-\sqrt{3}$	-1	$-\frac{\sqrt{3}}{3}$	0	—	0
$\cot\alpha$	—	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0	$-\frac{\sqrt{3}}{3}$	-1	$-\sqrt{3}$	—	0	—

Keltirish formulalari («Приведения») va Teskari trigonometrik funksiyalar

Funksiya	$-\alpha$	$\frac{\pi}{2}-\alpha$	$\frac{\pi}{2}+\alpha$	$\pi-\alpha$	$\pi+\alpha$	$\frac{3\pi}{2}-\alpha$	$\frac{3\pi}{2}+\alpha$	$2\pi-\alpha$
$\sin$	$-\sin\alpha$	$+\cos\alpha$	$+\cos\alpha$	$+\sin\alpha$	$-\sin\alpha$	$-\cos\alpha$	$-\cos\alpha$	$-\sin\alpha$
$\cos$	$+\cos\alpha$	$+\sin\alpha$	$-\sin\alpha$	$-\cos\alpha$	$-\cos\alpha$	$-\sin\alpha$	$+\sin\alpha$	$+\cos\alpha$
$\tan$	$-\tan\alpha$	$+\cot\alpha$	$-\cot\alpha$	$-\tan\alpha$	$+\tan\alpha$	$+\cot\alpha$	$-\cot\alpha$	$-\tan\alpha$
$\cot$	$-\cot\alpha$	$+\tan\alpha$	$-\tan\alpha$	$-\cot\alpha$	$+\cot\alpha$	$+\tan\alpha$	$-\tan\alpha$	$-\cot\alpha$

Teskari trigonometrik funksiyalarning manfiy argument xossalari: $\arcsin(-x)=-\arcsin x$, $\arccos(-x)=\pi-\arccos x$, $\arctan(-x)=-\arctan x$, $\operatorname{arccot}(-x)=\pi-\operatorname{arccot} x$. Shuningdek: $\arcsin x+\arccos x=\frac{\pi}{2}$ va $\arctan x+\operatorname{arccot} x=\frac{\pi}{2}$.

III BOB

Analitik Geometriyadagi Maxsus Mavzular va Tipovoy Sxemalar

Manba: `perehod_k_novomu_bazisu.html`, `lineinye_neravenstva.html`, `kak_nauchitsa_reshat_zadachi_po_geometrii.html` va `kak_privesti_uravnenie_linii_2_poryadka_k_kanonicheskomu_vidu.html`.

Mavzu 1. Yangi bazisga va yangi Dekart koordinatalar sistemasiga o'tish

1.1. Affin bazisni almashtirish va o'tish matritsasi

Tekislikda eski $(\vec{e}_1, \vec{e}_2)$ bazis va yangi $(\vec{e}'_1, \vec{e}'_2)$ bazis berilgan bo'lsin. Agar yangi bazis vektorlari eski bazis orqali quyidagicha yoyilsa:

$$\begin{cases} \vec{e}'_1 = \alpha_{11}\vec{e}_1 + \alpha_{21}\vec{e}_2 \\ \vec{e}'_2 = \alpha_{12}\vec{e}_1 + \alpha_{22}\vec{e}_2 \end{cases} \Longrightarrow C = \begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{pmatrix}$$

bu yerda C — **yangi bazisga o'tish matritsasi** (ustunlariga yangi bazis vektorlarining koordinatalari yoziladi!). Ixtiyoriy $\vec{x}$ vektorning eski koordinatalari $X = (x_1, x_2)^T$ va yangi koordinatalari $X' = (x'_1, x'_2)^T$ orasidagi bog'lanish:

$$X = C \cdot X' \quad \Longleftrightarrow \quad X' = C^{-1} \cdot X$$

1.2. To'g'ri burchakli Dekart koordinatalar sistemasini ko'chirish va burish

Agar koordinata boshi $O(0,0)$ nuqtadan $O'(x_0, y_0)$ nuqtaga parallel ko'chirilsa va o'qlar α burchakka soat miliga teskari yo'nalishda burilsa, nuqtaning eski (x, y) va yangi (x', y') koordinatalari quyidagi formulalar bilan bog'lanadi:

$$\begin{cases} x = x'\cos\alpha - y'\sin\alpha + x_0 \\ y = x'\sin\alpha + y'\cos\alpha + y_0 \end{cases} \Longleftrightarrow \begin{cases} x' = (x - x_0)\cos\alpha + (y - y_0)\sin\alpha \\ y' = -(x - x_0)\sin\alpha + (y - y_0)\cos\alpha \end{cases}$$

📝 **Namuna misol (Mathprofi.ru): Bazisga yoyish**

Masala: $\vec{a} = (1, 2)$, $\vec{b} = (3, -1)$ vektorlar bazis tashkil qilishini ko'rsating va $\vec{c} = (7, 0)$ vektorni shu bazis bo'yicha yoying.

Yechish: $\Delta = \begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} = -1 - 6 = -7 \neq 0 \Longrightarrow$ bazis tashkil qiladi. $\vec{c} = x'\vec{a} + y'\vec{b}$ dan sistema tuzamiz:

$\begin{cases} x' + 3y' = 7 \\ 2x' - y' = 0 \end{cases} \Longrightarrow x' = 1,\ y' = 2$. Demak, $\vec{c} = 1 \cdot \vec{a} + 2 \cdot \vec{b}$.

Mavzu 2. Tekislikdagi chiziqli tengsizliklar va sohalarni qurish (`lineinye_neravenstva.html`)

$Ax + By + C \geq 0$ (yoki > 0, ≤ 0, < 0) ko'rinishidagi chiziqli tengsizlik tekislikda **yarim tekislikni** aniqlaydi.

- **1-qadam:** $Ax + By + C = 0$ chegara to'g'ri chizig'ini yasaymiz (qat'iy tengsizlikda *uzlukli chiziq*, noqat'iy tengsizlikda *tutash chiziq*).
- **2-qadam (Sinov nuqtasi qoidasi):** To'g'ri chiziqda yotmaydigan bitta nuqtani, masalan $O(0,0)$ ni tengsizlikka qo'yamiz. Agar to'g'ri sonli tengsizlik chiqsa, $O(0,0)$ yotgan yarim tekislikni bo'yaymiz, aks holda qarama-qarshi yarim tekislikni olamiz.
- **3-qadam (Tezkor qoida $y \geq kx + b$):** Agar tengsizlik $y > kx + b$ ko'rinishga keltirilsa, to'g'ri chiziqdan **yuqoridagi** yarim tekislik olinadi; $y < kx + b$ bo'lsa, **pastdagi** yarim tekislik olinadi!

Mavzu 3. Tekislikda Uchburchakning Barcha Elementlarini Topish Tipovoy Sxemasi (`kak_nauchitsa_reshat_zadachi_po_geometrii.html`)

Universitet imtihonlarida va mustaqil ishlarda $A(x_A, y_A)$, $B(x_B, y_B)$, $C(x_C, y_C)$ uchlari berilgan $\triangle ABC$ uchun so'raladigan barcha 12 ta savolning yagona yechish jadvali:

№	Topilishi kerak bo'lgan element	Ishchi formula va algoritm
1	AB **tomon uzunligi**	$\vec{AB} = (x_B - x_A;\ y_B - y_A), \quad \lvert AB\rvert = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$

№	Topilishi kerak bo'lgan element	Ishchi formula va algoritm
2	AB **tomon tenglamasi**	$\frac{x-x_A}{x_B-x_A}=\frac{y-y_A}{y_B-y_A} \implies A_1x+B_1y+C_1=0, \quad k_{AB}=\frac{y_B-y_A}{x_B-x_A}$
3	$\angle A$ **ichki burchagi**	**1-usul:** $\cos A=\frac{\vec{AB}\cdot\vec{AC}}{\lvert\vec{AB}\rvert\cdot\lvert\vec{AC}\rvert}$. **2-usul:** $\tan A=\left\lvert\frac{k_{AC}-k_{AB}}{1+k_{AB}k_{AC}}\right\rvert$
4	CH **balandlik tenglamasi**	AB ning yo'naltiruvchi vektori $\vec{AB}$ — bu CH uchun **normal vektor**: $x_{AB}(x-x_C)+y_{AB}(y-y_C)=0$ (yoki $k_{CH}=-1/k_{AB}$)
5	CH **balandlik uzunligi**	**1-usul:** C nuqtadan $AB: A_1x+B_1y+C_1=0$ gacha masofa: $h_C=\frac{\lvert A_1x_C+B_1y_C+C_1\rvert}{\sqrt{A_1^2+B_1^2}}$. **2-usul:** $h_C=\frac{2S_{\triangle ABC}}{\lvert AB\rvert}$
6	AM **mediana tenglamasi**	BC o'rtasi $M\left(\frac{x_B+x_C}{2};\frac{y_B+y_C}{2}\right)$ ni topib, A va M nuqtalardan to'g'ri chiziq o'tkaziladi
7	**Og'irlik markazi** G **(Medianalar kesishmasi)**	$G\left(\frac{x_A+x_B+x_C}{3};\ \frac{y_A+y_B+y_C}{3}\right)$
8	**Ortomarkaz** H **(Balandliklar kesishmasi)**	Ikkita balandlik (CH va AH_1) tenglamalaridan tuzilgan chiziqli tenglamalar sistemasining yechimi
9	AL **ichki bissektrisa tenglamasi**	Birlik vektorlar yig'indisi $\vec{s}=\frac{\vec{AB}}{\lvert\vec{AB}\rvert}+\frac{\vec{AC}}{\lvert\vec{AC}\rvert}$ bissektrisaning yo'naltiruvchi vektori bo'ladi!
10	**Tashqi chizilgan aylana markazi** O_R	AB va BC tomonlarning o'rta perpendikulyarlari kesishish nuqtasi
11	**Uchburchak yuzi** $S_{\triangle ABC}$	$S=\frac{1}{2}\left\lvert\begin{vmatrix}x_B-x_A & y_B-y_A\\ x_C-x_A & y_C-y_A\end{vmatrix}\right\rvert$
12	C **nuqtaning** AB **ga nisbatan simmetrik** C' **nuqtasi**	AB va CH kesishmasidan $H(x_H,y_H)$ topiladi, so'ng $x_{C'}=2x_H-x_C,\ y_{C'}=2y_H-y_C$

Mavzu 4. Ikkinchi tartibli chiziq tenglamasini kanonik ko'rinishga keltirish

(`kak_privesti_uravnenie_linii_2_poryadka_k_kanonicheskomu_vidu.html`)

Ikkinchi tartibli chiziqning umumiy tenglamasi berilgan bo'lsin:

$$Ax^2+2Bxy+Cy^2+2Dx+2Ey+F=0$$

1-holat: $B=0$ **(**xy **ko'paytma qatnashmagan, faqat parallel ko'chirish)**

x va y bo'yicha to'la kvadratlarni ajratamiz: $A(x-x_0)^2+C(y-y_0)^2=F'$. Yangi koordinatalar $x'=x-x_0,\ y'=y-y_0$ kiritilib, ellips, giperbola yoki parabola kanonik tenglamasi olinadi.

2-holat: $B\neq 0$ **(**xy **had qatnashgan — Koordinata o'qlarini burish va Invariantlar usuli)**

1-usul (O'qlarni α **burchakka burish):** Koordinata o'qlarini $\cot 2\alpha=\frac{A-C}{2B}$ burchakka burganda ($x=x'\cos\alpha-y'\sin\alpha,\ y=x'\sin\alpha+y'\cos\alpha$) yangi tenglamada $x'y'$ had yo'qoladi!

2-usul (Ortogonal Invariantlar usuli): Koordinatalarni burish va ko'chirishda o'zgarmaydigan 3 ta miqdor (invariantlar) hisoblanadi:

$$I_1=A+C, \qquad I_2=\begin{vmatrix}A & B\\ B & C\end{vmatrix}=AC-B^2, \qquad I_3=\begin{vmatrix}A & B & D\\ B & C & E\\ D & E & F\end{vmatrix}$$

- Agar $I_2\neq 0$ (Markaziy chiziq: $I_2>0$ — ellips, $I_2<0$ — giperbola) bo'lsa, kanonik tenglama:

$$\lambda_1(x'')^2+\lambda_2(y'')^2+\frac{I_3}{I_2}=0$$

bu yerda λ_1,λ_2 — $\lambda^2-I_1\lambda+I_2=0$ xarakteristik tenglamaning ildizlari!

- Agar $I_2 = 0$ va $I_3 \neq 0$ (Parabola turi) bo'lsa, kanonik tenglama:

$$I_1(y'')^2 \pm 2\sqrt{-\frac{I_3}{I_1}}x'' = 0$$

📝 **Namuna misol:** $5x^2 + 4xy + 8y^2 - 32x - 56y + 80 = 0$ **chiziqni kanonik shaklga keltiring**

Yechish: Bu yerda $A = 5,\ B = 2,\ C = 8,\ D = -16,\ E = -28,\ F = 80$.

1) Invariantlarni topamiz: $I_1 = 5 + 8 = 13$, $I_2 = 5 \cdot 8 - 2^2 = 36 > 0$ (ellips turi).

2) $I_3 = \begin{vmatrix} 5 & 2 & -16 \\ 2 & 8 & -28 \\ -16 & -28 & 80 \end{vmatrix} = -1296$. Demak, $\frac{I_3}{I_2} = \frac{-1296}{36} = -36$.

3) Xarakteristik tenglama: $\lambda^2 - 13\lambda + 36 = 0 \implies \lambda_1 = 4,\ \lambda_2 = 9$.

4) Kanonik tenglama: $4(x'')^2 + 9(y'')^2 - 36 = 0 \implies \frac{(x'')^2}{9} + \frac{(y'')^2}{4} = 1$ (yarim o'qlari $a = 3, b = 2$ bo'lgan ellips!).

IV BOB

Oliy va Chiziqli Algebradagi Maxsus Mavzular

Manba: `reshenie_systemy_pri_razlichnyh_sposobah_vybora_bazisa.html`, `metod_ortogonalnogo_preobrazovaniya_kf.html` va `vyrazhenija_uravnenija_i_sistemy_s_kompleksnymi_chislami.html`.

Mavzu 5. Chiziqli tenglamalar sistemasini turli bazislarda yechish (Barcha tayanch yechimlarni topish)

Faraz qilaylik, m ta tenglama va n ta noma'lumli ($m < n$) chiziqli tenglamalar sistemasining rangi $r = m$ bo'lsin. U holda n ta o'zgaruvchidan r tasini **bazis o'zgaruvchilar**, qolgan $n - r$ tasini esa **erkin o'zgaruvchilar** deb tanlashning jami:

$$N \le C_n^r = \frac{n!}{r!(n-r)!}$$

ta usuli mavjud (agar tanlangan r ta ustundan tuzilgan determinant $\Delta \neq 0$ bo'lsa). Erkin o'zgaruvchilarni 0 ga tenglashtirganda hosil bo'ladigan xususiy yechim **bazis yechim** deb ataladi. Barcha komponentlari nomanfiy ($x_i \ge 0$) bo'lgan bazis yechim esa **tayanch yechim (опорное решение)** deyiladi (chiziqli dasturlashning asosi!).

📝 **Namuna misol: Barcha bazis yechimlarni topish**

Masala: $\begin{cases} x_1 + 2x_2 - x_3 = 4 \\ 2x_1 - x_2 + 3x_3 = 3 \end{cases}$ sistemaning barcha bazis yechimlarini toping.

Yechish: Bu yerda $n = 3, r = 2$, jami $C_3^2 = 3$ ta bazis juftligi bor:

- **1-bazis** (x_1, x_2)**, erkin** $x_3 = 0$**:** $\begin{cases} x_1 + 2x_2 = 4 \\ 2x_1 - x_2 = 3 \end{cases} \Longrightarrow x_1 = 2,\ x_2 = 1 \Longrightarrow X^{(1)} = (2;\ 1;\ 0)$ (tayanch yechim).
- **2-bazis** (x_1, x_3)**, erkin** $x_2 = 0$**:** $\begin{cases} x_1 - x_3 = 4 \\ 2x_1 + 3x_3 = 3 \end{cases} \Longrightarrow x_1 = 3,\ x_3 = -1 \Longrightarrow X^{(2)} = (3;\ 0;\ -1)$ (bazis yechim).
- **3-bazis** (x_2, x_3)**, erkin** $x_1 = 0$**:** $\begin{cases} 2x_2 - x_3 = 4 \\ -x_2 + 3x_3 = 3 \end{cases} \Longrightarrow x_2 = 3,\ x_3 = 2 \Longrightarrow X^{(3)} = (0;\ 3;\ 2)$ (tayanch yechim).

Mavzu 6. Kvadratik formani ortogonal almashtirish yordamida kanonik shaklga keltirish (`metod_ortogonalnogo_preobrazovaniya_kf.html`)

$Q(x_1, x_2, x_3) = X^T A X$ kvadratik formani ortogonal almashtirish $X = QY$ ($Q^T Q = E$, $\det Q = 1$) yordamida kanonik shaklga keltirishning **5 bosqichli algoritmi**:

1. Kvadratik formaning simmetrik matritsasi A ni yozamiz (bosh diagonalda kvadratlar koeffitsiyentlari, nodiagonalda esa ikkilangan ko'paytmalar koeffitsiyentining **yarmi**).
2. $\det(A - \lambda E) = 0$ xarakteristik tenglamadan xos sonlar $\lambda_1, \lambda_2, \lambda_3$ ni topamiz. (Ularning barchasi haqiqiy!).
3. Har bir λ_i uchun $(A - \lambda_i E)\vec{v}_i = \vec{0}$ sistemadan xos vektorlarni topamiz (turli λ_i larga mos xos vektorlar o'zaro ortogonal bo'ladi).
4. Har bir xos vektorni o'zining uzunligiga bo'lib **ortonormallaymiz**: $\vec{e}_i = \dfrac{\vec{v}_i}{|\vec{v}_i|}$ va ularni ustun qilib $Q = (\vec{e}_1 \mid \vec{e}_2 \mid \vec{e}_3)$ ortogonal matritsani tuzamiz.
5. Yangi koordinatalarda kvadratik formaning kanonik ko'rinishi bevosita yoziladi:

$$Q(y_1, y_2, y_3) = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$$

📝 **Namuna misol:** $Q(x_1,x_2)=5x_1^2+4x_1x_2+8x_2^2$ **ni ortogonal almashtirish bilan kanonik shaklga keltiring**

Yechish: 1) Matritsa: $A=\begin{pmatrix}5&2\\2&8\end{pmatrix}$. 2) $\begin{vmatrix}5-\lambda&2\\2&8-\lambda\end{vmatrix}=\lambda^2-13\lambda+36=0\implies\lambda_1=4,\ \lambda_2=9.$

3) $\lambda_1=4$ uchun: $x_1+2x_2=0\implies\vec{v}_1=(2,-1)^T\implies\vec{e}_1=\left(\frac{2}{\sqrt5},-\frac{1}{\sqrt5}\right)^T.$

4) $\lambda_2=9$ uchun: $-4x_1+2x_2=0\implies\vec{v}_2=(1,2)^T\implies\vec{e}_2=\left(\frac{1}{\sqrt5},\frac{2}{\sqrt5}\right)^T.$

5) Ortogonal almashtirish matritsasi $Q=\frac{1}{\sqrt5}\begin{pmatrix}2&1\\-1&2\end{pmatrix}$ va kanonik ko'rinishi: $Q(y_1,y_2)=4y_1^2+9y_2^2.$

Mavzu 7. Kompleks sonli tenglamalar va chiziqli sistemalar

(`vyrazhenija_uravnenija_i_sistemy_s_kompleksnymi_chislami.html`)

Kompleks koeffitsiyentli 2×2 chiziqli tenglamalar sistemasi Kramer qoidasi bilan aynan haqiqiy sonlardagi kabi yechiladi, faqat natijada maxrajdagi kompleks sonning qo'shmasiga ko'paytirib $a+bi$ algebraik shaklga keltiriladi.

📝 **Namuna misol (Mathprofi.ru): Kompleks ChTS**

Masala: $\begin{cases}(1+i)z_1-2z_2=3+i\\iz_1+(1-i)z_2=2\end{cases}$ sistemani Kramer usulida yeching.

Yechish: $\Delta=\begin{vmatrix}1+i&-2\\i&1-i\end{vmatrix}=(1+i)(1-i)-(-2i)=2+2i.$

$\Delta_1=\begin{vmatrix}3+i&-2\\2&1-i\end{vmatrix}=(3+i)(1-i)+4=(4-2i)+4=8-2i.$

$\Delta_2=\begin{vmatrix}1+i&3+i\\i&2\end{vmatrix}=2+2i-(3i-1)=3-i.$

$z_1=\frac{8-2i}{2+2i}=\frac{(4-i)(1-i)}{1^2+1^2}=\frac{3-5i}{2}=1.5-2.5i,\qquad z_2=\frac{3-i}{2+2i}=\frac{(3-i)(2-2i)}{8}=\frac{4-8i}{8}=0.5-i.$

V BOB

5 ta Ajoyib Limit, Ekvivalentliklar va Yuqori Qiyinlikdagi Limitlar

Manba: `zamechatelnye_predely.pdf`, `slozhnye_predely.html` va `predel_s_pomoshju_riada.html`.

Mavzu 8. Mathprofi.ru ning 5 ta Ajoyib Limiti va 14 ta Ekvivalentlik Jadvali

Ko'pgina darsliklarda faqat 2 ta ajoyib limit berilsa, `mathprofi.ru` ning maxsus ma'lumotnomasida amaliyotda eng ko'p ishlatiladigan **5 ta ajoyib limit** va ulardan kelib chiqadigan cheksiz kichik ekvivalentliklar ($\alpha(x) \to 0$ da) jamlangan:

№	Ajoyib limit nomi va formulasi	Umumlashgan shakli ($\alpha(x) \to 0$)	Ekvivalentlik ($\alpha \to 0$)
1	**Birinchi ajoyib limit:** $\lim_{x\to 0} \frac{\sin x}{x} = 1$	$\lim_{\alpha\to 0} \frac{\sin\alpha}{\alpha} = \lim_{\alpha\to 0} \frac{\tan\alpha}{\alpha} = \lim_{\alpha\to 0} \frac{\arcsin\alpha}{\alpha} = \lim_{\alpha\to 0} \frac{\arctan\alpha}{\alpha} = 1$	$\sin\alpha \sim \alpha,\ \tan\alpha \sim \alpha$ $\arcsin\alpha \sim \alpha,\ \arctan\alpha \sim \alpha$ $1-\cos\alpha \sim \frac{\alpha^2}{2}$
2	**Ikkinchi ajoyib limit:** $\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x = e$	$\lim_{\alpha\to 0}(1+\alpha)^{\frac{1}{\alpha}} = e, \qquad \lim_{x\to a} u(x)^{v(x)} = e^{\lim_{x\to a}(u(x)-1)v(x)}$	—
3	**Uchinchi ajoyib limit (Logarifmik):** $\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$	$\lim_{\alpha\to 0} \frac{\ln(1+\alpha)}{\alpha} = 1, \qquad \lim_{\alpha\to 0} \frac{\log_a(1+\alpha)}{\alpha} = \frac{1}{\ln a}$	$\ln(1+\alpha) \sim \alpha$ $\log_a(1+\alpha) \sim \frac{\alpha}{\ln a}$
4	**To'rtinchi ajoyib limit (Ko'rsatkichli):** $\lim_{x\to 0} \frac{e^x-1}{x} = 1$	$\lim_{\alpha\to 0} \frac{e^\alpha-1}{\alpha} = 1, \qquad \lim_{\alpha\to 0} \frac{a^\alpha-1}{\alpha} = \ln a$	$e^\alpha - 1 \sim \alpha$ $a^\alpha - 1 \sim \alpha\ln a$
5	**Beshinchi ajoyib limit (Darajali-binomial):** $\lim_{x\to 0} \frac{(1+x)^\mu-1}{x} = \mu$	$\lim_{\alpha\to 0} \frac{(1+\alpha)^\mu-1}{\alpha} = \mu, \qquad \lim_{\alpha\to 0} \frac{\sqrt[n]{1+\alpha}-1}{\alpha} = \frac{1}{n}$	$(1+\alpha)^\mu - 1 \sim \mu\alpha$ $\sqrt{1+\alpha}-1 \sim \frac{\alpha}{2}$

8.1. Yuqori qiyinlikdagi limitlarda o'zgaruvchini siljitish usuli ($t = x - x_0$)

Agar $x \to x_0$ ($x_0 \neq 0$, masalan $x \to \pi$, $x \to \frac{\pi}{2}$, $x \to a$) da trigonometrik yoki logarifmik $\left[\frac{0}{0}\right]$ aniqmaslik hosil bo'lsa, **har doim** $t = x - x_0 \implies x = x_0 + t$ **($t \to 0$)** almashtirish kiritiladi va keltirish formulalari yordamida 1-ajoyib limitga keltiriladi!

Namuna misol (Mathprofi.ru — slozhnye_predely.html)

Masala: $\lim_{x\to\pi/4} \frac{\cos x - \sin x}{\cos 2x}$ limitni 2 xil usulda hisoblang.

1-usul (Trigonometrik ko'paytuvchiga ajratish): $\cos 2x = \cos^2 x - \sin^2 x = (\cos x - \sin x)(\cos x + \sin x)$ ekanligidan:

$$\lim_{x\to\pi/4} \frac{\cos x - \sin x}{(\cos x - \sin x)(\cos x + \sin x)} = \lim_{x\to\pi/4} \frac{1}{\cos x + \sin x} = \frac{1}{\frac{\sqrt2}{2}+\frac{\sqrt2}{2}} = \frac{1}{\sqrt2} = \frac{\sqrt2}{2}$$

2-usul ($t = x - \pi/4$ almashtirish): $x = t + \frac{\pi}{4},\ t \to 0$. U holda $\cos\left(t+\frac{\pi}{4}\right) - \sin\left(t+\frac{\pi}{4}\right) = -\sqrt2\sin t$, maxrajda esa $\cos\left(2t+\frac{\pi}{2}\right) = -\sin 2t$. Limit: $\lim_{t\to 0} \frac{-\sqrt2\sin t}{-\sin 2t} = \lim_{t\to 0} \frac{\sqrt2 \cdot t}{2t} = \frac{\sqrt2}{2}$.

Mavzu 9. Makloren qatoriga yoyish yordamida murakkab limitlarni hisoblash (`predel_s_pomoshju_riada.html`)

Yig'indi yoki ayirmada birinchi tartibli cheksiz kichik hadlar o'zaro qisqarib ketganda (masalan, $\sin x - x$, $e^x - 1 - x$, $\cos x - 1 + \frac{x^2}{2}$), oddiy ekvivalentlik ($\sin x \sim x$) **ishlamaydi**, Lopital qoidasini esa 3–5 marta ketma-ket qo'llashga to'g'ri keladi. Bunday hollarda funksiyalarni Makloren qatorining dastlabki 2–3 ta hadiga yoyish limitni **bir qatorda** yechib beradi!

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + o(x^3), \qquad \sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5), \qquad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3), \qquad \arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} + o(x^5), \qquad \tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^5)$$

📝 **Namuna misol (Mathprofi.ru — predel_s_pomoshju_riada.html)**

Masala: $\lim_{x\to 0} \dfrac{e^x - e^{-x} - 2x}{x - \sin x}$ limitni Makloren qatori yordamida hisoblang.

Yechish: Surat va maxrajni x^3 hadgacha Makloren qatoriga yoyamiz:

$$e^x - e^{-x} - 2x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots\right) - \left(1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots\right) - 2x = \frac{2x^3}{6} + o(x^3) = \frac{x^3}{3} + o(x^3)$$

$$x - \sin x = x - \left(x - \frac{x^3}{6} + o(x^3)\right) = \frac{x^3}{6} + o(x^3)$$

$$\lim_{x\to 0} \frac{\frac{x^3}{3} + o(x^3)}{\frac{x^3}{6} + o(x^3)} = \frac{1/3}{1/6} = 2$$

VI BOB

Hosilalar Jadvali va Funksiyani To'liq Tekshirish Shpargalkasi

Manba: `tablica_proizvodnyh.pdf` va `shema_issledovaniya_funkcii.pdf`.

Mavzu 10. Oddiy $f(x)$ va Murakkab $f(u)$ Funksiyalar Hosilalarining Qiyosiy Jadvali

№	Oddiy funksiya $y = f(x)$	Murakkab funksiya $y = f(u),\ u = u(x)$	№	Oddiy funksiya $y = f(x)$	Murakkab funksiya $y = f(u),\ u = u(x)$
1	$(C)' = 0, \quad (x)' = 1$	$(C \cdot u)' = C \cdot u'$	10	$(\tan x)' = \dfrac{1}{\cos^2 x}$	$(\tan u)' = \dfrac{u'}{\cos^2 u}$
2	$(x^n)' = nx^{n-1}$	$(u^n)' = nu^{n-1} \cdot u'$	11	$(\cot x)' = -\dfrac{1}{\sin^2 x}$	$(\cot u)' = -\dfrac{u'}{\sin^2 u}$
3	$(\sqrt{x})' = \dfrac{1}{2\sqrt{x}}$	$(\sqrt{u})' = \dfrac{u'}{2\sqrt{u}}$	12	$(\arcsin x)' = \dfrac{1}{\sqrt{1-x^2}}$	$(\arcsin u)' = \dfrac{u'}{\sqrt{1-u^2}}$
4	$\left(\dfrac{1}{x}\right)' = -\dfrac{1}{x^2}$	$\left(\dfrac{1}{u}\right)' = -\dfrac{u'}{u^2}$	13	$(\arccos x)' = -\dfrac{1}{\sqrt{1-x^2}}$	$(\arccos u)' = -\dfrac{u'}{\sqrt{1-u^2}}$
5	$(e^x)' = e^x$	$(e^u)' = e^u \cdot u'$	14	$(\arctan x)' = \dfrac{1}{1+x^2}$	$(\arctan u)' = \dfrac{u'}{1+u^2}$
6	$(a^x)' = a^x \ln a$	$(a^u)' = a^u \ln a \cdot u'$	15	$(\operatorname{arccot} x)' = -\dfrac{1}{1+x^2}$	$(\operatorname{arccot} u)' = -\dfrac{u'}{1+u^2}$
7	$(\ln x)' = \dfrac{1}{x}$	$(\ln u)' = \dfrac{u'}{u}$	16	$(\sinh x)' = \cosh x$	$(\sinh u)' = \cosh u \cdot u'$
8	$(\sin x)' = \cos x$	$(\sin u)' = \cos u \cdot u'$	17	$(\cosh x)' = \sinh x$	$(\cosh u)' = \sinh u \cdot u'$
9	$(\cos x)' = -\sin x$	$(\cos u)' = -\sin u \cdot u'$	18	$(\tanh x)' = \dfrac{1}{\cosh^2 x}$	$(u^v)' = u^v \left(v' \ln u + \dfrac{vu'}{u}\right)$

10.1. Funksiyani to'liq tekshirish va grafigini yasashning 8 bosqichli algoritmi (`shema_issledovaniya_funkcii.pdf`)

1. **Aniqlanish sohasi $D(y)$ va uzluksizlik:** Maxraj $\neq 0$, juft ildiz osti ≥ 0, logarifm ichi > 0. Chegara (uzilish) nuqtalarida bir tomonlama limitlar $\lim_{x \to x_0 \pm 0} f(x)$ topiladi.
2. **Juftlik, toqlik va davriylik:** $f(-x) = f(x)$ — juft (Oy ga nisbatan simmetrik), $f(-x) = -f(x)$ — toq ($O(0,0)$ ga nisbatan simmetrik).
3. **Koordinata o'qlari bilan kesishish va ishora doimiylik oraliqlari:** $x = 0 \implies y = f(0)$; $f(x) = 0 \implies x_i$ nollari va intervallar usuli bilan $f(x) > 0,\ f(x) < 0$ oraliqlar.
4. **Asimptotalar:**
   - *Vertikal:* $\lim_{x \to a \pm 0} f(x) = \pm\infty \implies x = a$.
   - *Og'ma ($y = kx + b$):* $k = \lim_{x \to \pm\infty} \dfrac{f(x)}{x}, \quad b = \lim_{x \to \pm\infty} [f(x) - kx]$ ($k = 0$ bo'lsa gorizontal asimptota $y = b$).
5. **Monotonlik va Ekstremumlar (y' bo'yicha):** $f'(x) = 0$ yoki $f'(x)$ mavjud bo'lmagan kritik nuqtalar topiladi. $f'(x) > 0 \implies \uparrow$, $f'(x) < 0 \implies \downarrow$. Ishora $+ \to -$ bo'lsa $x_{\max}$, $- \to +$ bo'lsa $x_{\min}$.
6. **Qavariqlik, botiqlik va Egilish nuqtalari (y'' bo'yicha):** $f''(x) = 0$ nuqtalar topiladi. $f''(x) > 0 \implies \cup$ (botiq / pastga qavariq), $f''(x) < 0 \implies \cap$ (qavariq / yuqoriga qavariq). $f''(x)$ ishorasi almashgan nuqta — **egilish nuqtasi**.
7. **Qo'shimcha (nazorat) nuqtalar va Yig'ma jadval tuzish.**
8. **Asimptotalar va tayanch nuqtalar asosida funksiya grafigini chizish.**

VII BOB

Ko'p O'zgaruvchili Funksiyalarning Maxsus Mavzulari

Manba: `predel_funkcii_dvuh_peremennyh.html`, `povtornye_predely.html`, `nepreryvnost_funkcii_dvuh_peremennyh.html`, `kak_proverit_udovletvorjaet_li_funkcija_uravneniju.html` va `metod_naimenshih_kvadratov.html`.

Mavzu 11. Ikki o'zgaruvchili funksiyaning ikki karrali (umumiy) limiti va takroriy limitlari

11.1. Ikki karrali (umumiy) limit va yo'nalishga bog'liqlik

$\lim_{x\to x_0,\ y\to y_0} f(x,y) = A$ umumiy limit mavjud bo'lishi uchun $M(x,y)$ nuqta $M_0(x_0,y_0)$ nuqtaga **ixtiyoriy yo'nalish (egri chiziq) bo'ylab** yaqinlashganda funksiya qiymati bir xil A songa intilishi shart!

- **Limit mavjud emasligini isbotlash usuli ($y = kx$ nurlar dastasi):** Agar $(x,y) \to (0,0)$ da $y = kx$ (yoki $y = kx^{\alpha}$) almashtirish qo'yganda limit natijasi k burchak koeffitsiyentiga bog'liq chiqsa, **umumiy limit mavjud emas!**
- **Qutb koordinatalari usuli:** $(x,y) \to (0,0)$ da $x = r\cos\varphi,\ y = r\sin\varphi\ (r \to 0)$ almashtirish kiritiladi. Agar $\lim_{r\to 0} f(r\cos\varphi, r\sin\varphi) = A$ burchak φ ga bog'liq bo'lmasa (tekis yaqinlashsa), umumiy limit A ga teng!

11.2. Takroriy (ketma-ket) limitlar (`povtornye_predely.html`)

Ikki o'zgaruvchili $f(x,y)$ funksiya uchun $M_0(x_0,y_0)$ nuqtada ikkita **takroriy limit** mavjud:

$$L_1 = \lim_{x\to x_0}\left(\lim_{y\to y_0} f(x,y)\right), \qquad L_2 = \lim_{y\to y_0}\left(\lim_{x\to x_0} f(x,y)\right)$$

⚠️ **Muhim nazariy xulosa**

Agar $L_1 \neq L_2$ bo'lsa, umumiy limit $\lim_{x\to x_0,\ y\to y_0} f(x,y)$ **aniq mavjud emas**. Ammo $L_1 = L_2$ bo'lishidan umumiy limit mavjudligi kelib chiqmaydi!

📝 **Namuna misol (Mathprofi.ru): Umumiy va takroriy limitlarni solishtirish**

Masala: $f(x,y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ funksiyaning $(0,0)$ nuqtadagi takroriy limitlarini va umumiy limitini tekshiring.

Yechish: 1) Takroriy limitlar:

$$L_1 = \lim_{x\to 0}\left(\lim_{y\to 0}\frac{x^2-y^2}{x^2+y^2}\right) = \lim_{x\to 0}\frac{x^2-0}{x^2+0} = 1, \qquad L_2 = \lim_{y\to 0}\left(\lim_{x\to 0}\frac{x^2-y^2}{x^2+y^2}\right) = \lim_{y\to 0}\frac{0-y^2}{0+y^2} = -1$$

$L_1 = 1 \neq L_2 = -1$ bo'lgani uchun umumiy limit mavjud emas!

2) $y = kx$ yo'nalish bo'yicha tekshirsak ham: $\lim_{x\to 0}\dfrac{x^2 - k^2x^2}{x^2 + k^2x^2} = \dfrac{1-k^2}{1+k^2}$ (k ga bog'liq, demak umumiy limit yo'q).

Mavzu 12. Ikki o'zgaruvchili funksiya uzluksizligi va uzilish chiziqlari (`nepreryvnost_funkcii_dvuh_peremennyh.html`)

$z = f(x,y)$ funksiya $M_0(x_0,y_0)$ nuqtada uzluksiz deyiladi, agar $\lim_{x\to x_0,\ y\to y_0} f(x,y) = f(x_0,y_0)$ bo'lsa. Ikki o'zgaruvchili funksiyalarda uzilishlar **yakkalangan uzilish nuqtasi** (masalan, $z = \frac{1}{x^2+y^2}$ da $(0,0)$ nuqta) yoki **uzilish chiziqlari** (masalan, $z = \frac{1}{x-y}$ da $y = x$ to'g'ri chizig'i, $z = \frac{1}{x^2+y^2-4}$ da $x^2 + y^2 = 4$ aylana) ko'rinishida bo'ladi.

Mavzu 13. Funksiya berilgan xususiy hosilali tenglamani qanoatlantirishini tekshirish (`kak_proverit_udovletvorjaet_li_funkcija_uravneniju.html`)

Universitet nazorat ishlarida juda ko'p tushadigan tipovoy masala: berilgan $z = f(x,y)$ funksiya ko'rsatilgan xususiy hosilali tenglamani (masalan, Eyler tenglamasi $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = nz$ yoki Laplas tenglamasi $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$) qanoatlantirishini isbotlash.

📝 **Namuna misol (Mathprofi.ru)**

Masala: $z = y\ln(x^2 - y^2)$ funksiya $\frac{1}{x}\frac{\partial z}{\partial x} + \frac{1}{y}\frac{\partial z}{\partial y} = \frac{z}{y^2}$ tenglamani qanoatlantirishini tekshiring.

Yechish: 1) Xususiy hosilalarni topamiz:

$$\frac{\partial z}{\partial x} = y \cdot \frac{2x}{x^2 - y^2} = \frac{2xy}{x^2 - y^2}, \qquad \frac{\partial z}{\partial y} = 1 \cdot \ln(x^2 - y^2) + y \cdot \frac{-2y}{x^2 - y^2} = \ln(x^2 - y^2) - \frac{2y^2}{x^2 - y^2}$$

2) Tenglama chap qismiga qo'yamiz:

$$\frac{1}{x} \cdot \frac{2xy}{x^2 - y^2} + \frac{1}{y}\left(\ln(x^2 - y^2) - \frac{2y^2}{x^2 - y^2}\right) = \frac{2y}{x^2 - y^2} + \frac{\ln(x^2 - y^2)}{y} - \frac{2y}{x^2 - y^2} = \frac{y\ln(x^2 - y^2)}{y^2} = \frac{z}{y^2}$$

Ayniyat hosil bo'ldi, demak qanoatlantiradi!

Mavzu 14. Ikki o'zgaruvchili funksiya ekstremumi sifatida Eng Kichik Kvadratlar Usuli

(`metod_naimenshih_kvadratov.html`)

Tajribada $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ nuqtalar berilgan bo'lsin. Ularga eng yaqin o'tuvchi $y = ax + b$ to'g'ri chiziqning a va b parametrlarini topish uchun chetlanishlar kvadratlari yig'indisi (ikki o'zgaruvchili funksiya):

$$S(a, b) = \sum_{i=1}^{n} (ax_i + b - y_i)^2 \to \min$$

minimallashtiriladi. Ekstremumning zaruriy sharti $\frac{\partial S}{\partial a} = 0,\ \frac{\partial S}{\partial b} = 0$ dan **normal tenglamalar sistemasi** kelib chiqadi:

$$\begin{cases} a\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i \\ a\sum_{i=1}^{n} x_i + n \cdot b = \sum_{i=1}^{n} y_i \end{cases}$$

VIII BOB

Bir Karrali Integrallarning Maxsus Jadvallari va Yuqori Qiyinlikdagi Usullari

Manba: `tablica_integralov.pdf`, `integrirovanie_drobei.html`, `integraly_ot_trigonometricheskih_funkcij.html`, `slozhnye_integraly.html` va `metody_reshenija_opredelennyh_i_nesobstvennyh_integralov.html`.

Mavzu 15. Mathprofi.ru ning Kengaytirilgan Integrallar Jadvali va Differensial Ostiga Kiritish

№	Jadval integrali	№	Jadval integrali
1	$\int u^n du = \frac{u^{n+1}}{n+1} + C \ (n \neq -1)$	11	$\int \frac{du}{u^2+a^2} = \frac{1}{a}\arctan\frac{u}{a} + C$
2	$\int \frac{du}{u} = \ln\lvert u\rvert + C$	12	$\int \frac{du}{u^2-a^2} = \frac{1}{2a}\ln\left\lvert\frac{u-a}{u+a}\right\rvert + C$
3	$\int e^u du = e^u + C, \quad \int a^u du = \frac{a^u}{\ln a} + C$	13	$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\frac{u}{a} + C$
4	$\int \sin u\, du = -\cos u + C, \quad \int \cos u\, du = \sin u + C$	14	$\int \frac{du}{\sqrt{u^2 \pm a^2}} = \ln\left\lvert u + \sqrt{u^2 \pm a^2}\right\rvert + C$
5	$\int \frac{du}{\cos^2 u} = \tan u + C, \quad \int \frac{du}{\sin^2 u} = -\cot u + C$	15	$\int \sqrt{a^2-u^2}\, du = \frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\arcsin\frac{u}{a} + C$
6	$\int \tan u\, du = -\ln\lvert\cos u\rvert + C$	16	$\int \sqrt{u^2 \pm a^2}\, du = \frac{u}{2}\sqrt{u^2 \pm a^2} \pm \frac{a^2}{2}\ln\left\lvert u + \sqrt{u^2 \pm a^2}\right\rvert + C$
7	$\int \cot u\, du = \ln\lvert\sin u\rvert + C$	17	$\int \sinh u\, du = \cosh u + C, \quad \int \cosh u\, du = \sinh u + C$
8	$\int \frac{du}{\sin u} = \ln\left\lvert\tan\frac{u}{2}\right\rvert + C$	18	$\int e^{ax}\sin bx\, dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + C$
9	$\int \frac{du}{\cos u} = \ln\left\lvert\tan\left(\frac{u}{2}+\frac{\pi}{4}\right)\right\rvert + C$	19	$\int e^{ax}\cos bx\, dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2} + C$
10	$\int \ln u\, du = u\ln u - u + C$	20	**Differensial ostiga kiritish:** $f'(x)dx = d(f(x))$

Mavzu 16. Maxrajda kvadrat uchhad qatnashgan kasrlarni integrallash va Ostrogradskiy usuli (`integrirovanie_drobei.html`)

1) $\int \frac{Ax+B}{ax^2+bx+c}dx$ **va** $\int \frac{Ax+B}{\sqrt{ax^2+bx+c}}dx$ **turidagi integrallar:**
Suratda maxrajdagi kvadrat uchhadning hosilasi $(ax^2+bx+c)' = 2ax+b$ sun'iy ravishda ajratiladi:

$$Ax + B = \frac{A}{2a}(2ax+b) + \left(B - \frac{Ab}{2a}\right)$$

Natijada birinchi integral $\int \frac{d(ax^2+bx+c)}{ax^2+bx+c} = \ln\lvert ax^2+bx+c\rvert$ (yoki $2\sqrt{ax^2+bx+c}$) ga, ikkinchi integral esa maxrajda to'la kvadrat ajratish orqali $\arctan$ yoki «uzun logarifm»ga keladi.

2) Karrali ildizlarda Ostrogradskiy usuli: Agar ratsional kasr maxraji $Q(x)$ yuqori karrali ildizlarga ega bo'lsa, sodda kasrlarga yoymasdan **Ostrogradskiy formulasi** qo'llaniladi:

$$\int \frac{P(x)}{Q(x)}dx = \frac{P_1(x)}{Q_1(x)} + \int \frac{P_2(x)}{Q_2(x)}dx$$

bu yerda $Q_1(x) = \text{EKUB}(Q(x), Q'(x))$, $Q_2(x) = Q(x)/Q_1(x)$ (faqat oddiy ildizlarga ega ko'phad!).

Mavzu 17. Trigonometrik integrallarning barcha turlari xaritasi (`integraly_ot_trigonometricheskih_funkcij.html`)

Integral ko'rinishi	Qaysi almashtirish yoki formula qo'llaniladi?
$\int \sin^m x \cos^n x\, dx$ (m — toq)	Bitta $\sin x\, dx = -d(\cos x)$ ajratiladi, $\sin^{m-1} x = (1-\cos^2 x)^{\frac{m-1}{2}}$, $t = \cos x$.
$\int \sin^m x \cos^n x\, dx$ (n — toq)	Bitta $\cos x\, dx = d(\sin x)$ ajratiladi, $\cos^{n-1} x = (1-\sin^2 x)^{\frac{n-1}{2}}$, $t = \sin x$.
$\int \sin^m x \cos^n x\, dx$ (m, n — juft nomanfiy)	Darajani pasaytirish formulalari: $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$, $\sin x \cos x = \frac{1}{2}\sin 2x$.
$\int \tan^n x\, dx$ yoki $\int \cot^n x\, dx$	$\tan^2 x = \frac{1}{\cos^2 x} - 1$ ajratilib, daraja $n-2$ ga tushiriladi (yoki $t = \tan x,\ dx = \frac{dt}{1+t^2}$).
$\int R(\sin^2 x, \cos^2 x, \sin x \cos x, \tan x)dx$	$R(-\sin x, -\cos x) = R(\sin x, \cos x)$ bo'lsa: $t = \tan x$, $\sin^2 x = \frac{t^2}{1+t^2},\ \cos^2 x = \frac{1}{1+t^2},\ dx = \frac{dt}{1+t^2}$.
Ixtiyoriy $\int R(\sin x, \cos x)dx$	**Universal Veyershtrass almashtirishi:** $t = \tan\frac{x}{2}$, $\sin x = \frac{2t}{1+t^2},\ \cos x = \frac{1-t^2}{1+t^2},\ dx = \frac{2dt}{1+t^2}$.

Mavzu 18. Chebishevning binomial differensiallari (`slozhnye_integraly.html`)

$\int x^m(a+bx^n)^p\, dx$ ($m, n, p \in \mathbb{Q}$) ko'rinishidagi integral (binomial differensial) **Chebishev teoremasiga** ko'ra faqat quyidagi **3 ta holdagina** elementar funksiyalarda integrallanadi:

1. $p \in \mathbb{Z}$ **(butun son):** Agar $m = r_1/s_1, n = r_2/s_2$ kasr bo'lsa, $x = t^k$ ($k = \text{EKUK}(s_1, s_2)$) almashtirish kiritiladi.
2. $\dfrac{m+1}{n} \in \mathbb{Z}$ **(butun son):** $a + bx^n = t^s$ almashtirish kiritiladi (s — p kasrning maxraji).
3. $\dfrac{m+1}{n} + p \in \mathbb{Z}$ **(butun son):** $ax^{-n} + b = t^s$ almashtirish kiritiladi (s — p kasrning maxraji).

Mavzu 19. Aniq va xosmas integrallarni hisoblashning effektiv usullari (`metody_reshenija_opredelennyh_i_nesobstvennyh_integralov.html`)

- **Simmetrik $[-a, a]$ oraliq xossasi:** Agar $f(x)$ toq bo'lsa, $\int_{-a}^{a} f(x)dx = 0$; agar $f(x)$ juft bo'lsa, $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$.
- **Ko'zgu almashtirishi:** $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$. Xususan, $\int_0^{\pi/2} f(\sin x)dx = \int_0^{\pi/2} f(\cos x)dx$.
- **Eylerning Gamma va Beta funksiyalari:** $\Gamma(p) = \int_0^{+\infty} x^{p-1}e^{-x}dx$, $\Gamma(n+1) = n!$, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, $\int_0^{+\infty} e^{-x^2}dx = \dfrac{\sqrt{\pi}}{2}$.

IX BOB

Differensial Tenglamalarning Maxsus Turlari va Xususiy Yechim Jadvali

Manba: `differencialnye_uravnenija_dopuskajushie_ponizhenie_poryadka.html`, `linejnye_diffury_vysshih_porjadkov.html`, `kak_podobrat_chastnoe_reshenie_dy.pdf` va `zadachi_s_diffurami.html`.

Mavzu 20. Tartibini pasaytirish mumkin bo'lgan yuqori tartibli differensial tenglamalar

`mathprofi.ru` da alohida ma'ruza sifatida ajratilgan **3 ta klassik tur**:

1. **1-tur:** $y^{(n)} = f(x)$ **(faqat x ga bog'liq).** Ketma-ket n marta integrallash orqali yechiladi:

$$y'' = 6x + 2 \implies y' = 3x^2 + 2x + C_1 \implies y = x^3 + x^2 + C_1 x + C_2$$

2. **2-tur:** $F(x, y', y'') = 0$ **(noma'lum funksiya y oshkor qatnashmagan).**
   Almashtirish: $y'(x) = p(x) \implies y''(x) = p'(x)$. Natijada $p(x)$ ga nisbatan 1-tartibli tenglama hosil bo'ladi!
3. **3-tur:** $F(y, y', y'') = 0$ **(erkli o'zgaruvchi x oshkor qatnashmagan — Avtonom tenglama).**
   Almashtirish: $y' = p(y)$ (p ni y ning funksiyasi deb qaraymiz!), u holda zanjir qoidasiga ko'ra:

$$y'' = \frac{dp}{dx} = \frac{dp}{dy} \cdot \frac{dy}{dx} = p \cdot \frac{dp}{dy} = p' \cdot p$$

📝 **Namuna misol (3-tur avtonom tenglama):** $y \cdot y'' - (y')^2 = 0$ **ni yeching**

Yechish: Tenglama x ni oshkor saqlamaydi. $y' = p(y),\ y'' = p\dfrac{dp}{dy}$ almashtirish kiritamiz:

$$y \cdot p\frac{dp}{dy} - p^2 = 0 \implies p\left(y\frac{dp}{dy} - p\right) = 0$$

1) $p = 0 \implies y' = 0 \implies y = C$.
2) $\dfrac{dp}{p} = \dfrac{dy}{y} \implies \ln|p| = \ln|C_1 y| \implies p = C_1 y \implies \dfrac{dy}{dx} = C_1 y \implies \ln|y| = C_1 x + \ln|C_2| \implies y = C_2 e^{C_1 x}$.

Mavzu 21. Yuqori tartibli ($n \geq 3$) o'zgarmas koeffitsiyentli chiziqli DT

(`linejnye_diffury_vysshih_porjadkov.html`)

$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y = 0$ tenglama uchun $k^n + a_1 k^{n-1} + \cdots + a_n = 0$ xarakteristik tenglama tuziladi (ratsional ildizlar Gorner sxemasi bilan topiladi):

- Har bir oddiy haqiqiy ildiz k_i ga $C_i e^{k_i x}$ mos keladi;
- m karrali haqiqiy ildiz k ga $(C_1 + C_2 x + \cdots + C_m x^{m-1})e^{kx}$ mos keladi;
- Oddiy kompleks-qo'shma ildizlar $\alpha \pm \beta i$ ga $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$ mos keladi.

Mavzu 22. Xususiy Yechim y_{xus} ni Tanlashning To'liq 12 Qatorli Mathprofi Jadvali

(`kak_podobrat_chastnoe_reshenie_dy.pdf`)

$ay'' + by' + cy = f(x)$ tenglamaning o'ng tomoni $f(x)$ ko'rinishiga va xarakteristik tenglama $ak^2 + bk + c = 0$ ildizlari (k_1, k_2) bilan ustma-ust tushish (rezonans) holatiga qarab xususiy yechim y_{xus} quyidagi jadval bo'yicha tanlanadi:

№	O'ng tomon $f(x)$ ko'rinishi	Nazorat soni	Rezonans sharti (k_1, k_2 bilan solishtirish)	Xususiy yechim y_{xus} shakli
1	$P_n(x)$ (n-darajali ko'phad)	$k = 0$	0 ildiz emas ($c \neq 0$) 0 oddiy ildiz ($c = 0, b \neq 0$) 0 ikki karrali ildiz ($b = c = 0$)	$Q_n(x)$ $x \cdot Q_n(x)$ $x^2 \cdot Q_n(x)$

№	O'ng tomon $f(x)$ ko'rinishi	Nazorat soni	Rezonans sharti (k_1, k_2 bilan solishtirish)	Xususiy yechim y_{xus} shakli
2	$ae^{\alpha x}$	$k = \alpha$	α ildiz emas α oddiy ildiz ($s = 1$) α ikki karrali ildiz ($s = 2$)	$Ae^{\alpha x}$ $Axe^{\alpha x}$ $Ax^2e^{\alpha x}$
3	$P_n(x)e^{\alpha x}$	$k = \alpha$	α ildiz emas ($s = 0$) α s-karrali ildiz ($s = 1$ yoki 2)	$Q_n(x)e^{\alpha x}$ $x^sQ_n(x)e^{\alpha x}$
4	$M\cos\beta x + N\sin\beta x$	$k = \pm\beta i$	$\pm\beta i$ ildiz emas $\pm\beta i$ ildiz bo'lsa	$A\cos\beta x + B\sin\beta x$ $x(A\cos\beta x + B\sin\beta x)$
5	$e^{\alpha x}(M\cos\beta x + N\sin\beta x)$	$k = \alpha \pm \beta i$	$\alpha \pm \beta i$ ildiz emas $\alpha \pm \beta i$ ildiz bo'lsa	$e^{\alpha x}(A\cos\beta x + B\sin\beta x)$ $xe^{\alpha x}(A\cos\beta x + B\sin\beta x)$
6	$[P_n(x)\cos\beta x + R_m(x)\sin\beta x]e^{\alpha x}$	$k = \alpha \pm \beta i$	$\alpha \pm \beta i$ s-karrali ($s = 0$ yoki 1)	$x^s[U_N(x)\cos\beta x + V_N(x)\sin\beta x]e^{\alpha x},\ N = \max(n, m)$

Izoh: $Q_n(x) = A_nx^n + A_{n-1}x^{n-1} + \cdots + A_1x + A_0$ — barcha darajalari to'liq yozilgan noma'lum koeffitsiyentli ko'phad (hatto $P_n(x) = x^2$ bo'lsa ham $Q_2(x) = Ax^2 + Bx + C$ olinadi!).

Mavzu 23. Geometrik va fizik masalalardan DT tuzish (`zadachi_s_diffurami.html`)

- **Geometrik masalalar:** $M(x, y)$ nuqtadagi urinmaning burchak koeffitsiyenti $k = y'(x)$, urinmaning Ox o'qidan kesgan nuqtasi $X_u = x - \dfrac{y}{y'}$, Oy o'qidan kesgan nuqtasi $Y_u = y - xy'$, suburinma uzunligi $|y/y'|$, subnormal uzunligi $|y \cdot y'|$.
- **Ortogonal trayektoriyalar:** Berilgan $F(x, y, C) = 0$ chiziqlar oilasini 90° burchak ostida kesib o'tuvchi chiziqlar oilasini topish uchun oilaning differensial tenglamasida y' o'rniga $-\dfrac{1}{y'}$ qo'yilib integrallanadi!

X BOB

Funksional Qatorlar, Tekis Yaqinlashish va Makloren Yoyilmalari Jadvali

Manba: `ravnomernaja_shodimost.html`, `slozhnye_funkcionalnye_ryady.html` va `tablica_razlozhenij_funkcij_v_ryady.pdf`.

Mavzu 24. Funksional qatorlarning tekis yaqinlashishi va Veyershtrass alomati

$\sum_{n=1}^{\infty} u_n(x)$ funksional qator X sohada $S(x)$ yig'indiga **tekis yaqinlashadi (равномерно сходится)** deyiladi, agar har qanday $\varepsilon > 0$ uchun x ga bog'liq bo'lmagan shunday $N(\varepsilon)$ nomer topilsaki, barcha $n > N(\varepsilon)$ va barcha $x \in X$ uchun $|R_n(x)| = |S(x) - S_n(x)| < \varepsilon$ bajarilsa (ya'ni $\lim_{n\to\infty} \sup_{x\in X} |R_n(x)| = 0$).

Veyershtrassning majorant alomati (Amaliyotdagi asosiy qurol)

Agar X sohadagi barcha x lar uchun $|u_n(x)| \le a_n$ $(n = 1, 2, \ldots)$ tengsizlik bajarilsa va $\sum_{n=1}^{\infty} a_n$ musbat hadli sonli qator (majorant qator) yaqinlashsa, u holda $\sum_{n=1}^{\infty} u_n(x)$ funksional qator X sohada **mutlaq va tekis yaqinlashadi!**

Namuna misol (Mathprofi.ru — ravnomernaja_shodimost.html)

Masala: $\sum_{n=1}^{\infty} \dfrac{\cos nx}{n^2 + x^2}$ qator $(-\infty, +\infty)$ da tekis yaqinlashishini isbotlang.

Yechish: Ixtiyoriy $x \in \mathbb{R}$ uchun $|\cos nx| \le 1$ va $n^2 + x^2 \ge n^2$ bo'lgani sababli:

$$\left|\frac{\cos nx}{n^2 + x^2}\right| \le \frac{1}{n^2} = a_n, \qquad \forall x \in (-\infty, +\infty)$$

$\sum_{n=1}^{\infty} \dfrac{1}{n^2}$ umumlashgan garmonik qator ($p = 2 > 1$) yaqinlashadi. Demak, Veyershtrass alomatiga ko'ra berilgan funksional qator butun $\mathbb{R}$ da tekis yaqinlashadi!

Mavzu 25. Makloren Darajali Qatorlariga Yoyishning To'liq Jadvali (`tablica_razlozhenij_funkcij_v_ryady.pdf`)

№	Funksiya $f(x)$	Makloren qatori yoyilmasi ($x_0 = 0$)	Yaqinlashish sohasi
1	e^x	$1 + \dfrac{x}{1!} + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots + \dfrac{x^n}{n!} + \cdots = \sum_{n=0}^{\infty} \dfrac{x^n}{n!}$	$(-\infty, +\infty)$
2	$\sin x$	$x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots + (-1)^n \dfrac{x^{2n+1}}{(2n+1)!} + \ldots$	$(-\infty, +\infty)$
3	$\cos x$	$1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \cdots + (-1)^n \dfrac{x^{2n}}{(2n)!} + \ldots$	$(-\infty, +\infty)$
4	$\sinh x$	$x + \dfrac{x^3}{3!} + \dfrac{x^5}{5!} + \cdots + \dfrac{x^{2n+1}}{(2n+1)!} + \ldots$	$(-\infty, +\infty)$
5	$\cosh x$	$1 + \dfrac{x^2}{2!} + \dfrac{x^4}{4!} + \cdots + \dfrac{x^{2n}}{(2n)!} + \ldots$	$(-\infty, +\infty)$
6	$\ln(1 + x)$	$x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \dfrac{x^4}{4} + \cdots + (-1)^{n-1} \dfrac{x^n}{n} + \ldots$	$(-1,\ 1]$
7	$\ln(1 - x)$	$-x - \dfrac{x^2}{2} - \dfrac{x^3}{3} - \dfrac{x^4}{4} - \cdots - \dfrac{x^n}{n} - \ldots$	$[-1,\ 1)$
8	$\arctan x$	$x - \dfrac{x^3}{3} + \dfrac{x^5}{5} - \cdots + (-1)^n \dfrac{x^{2n+1}}{2n+1} + \ldots$	$[-1,\ 1]$
9	$\arcsin x$	$x + \dfrac{1}{2}\dfrac{x^3}{3} + \dfrac{1 \cdot 3}{2 \cdot 4}\dfrac{x^5}{5} + \cdots + \dfrac{(2n-1)!!}{(2n)!!}\dfrac{x^{2n+1}}{2n+1} + \ldots$	$[-1,\ 1]$
10	$(1 + x)^{\alpha}$ (Binomial)	$1 + \alpha x + \dfrac{\alpha(\alpha-1)}{2!}x^2 + \cdots + \dfrac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + \ldots$	$(-1,\ 1)$
11	$\dfrac{1}{1 - x}$	$1 + x + x^2 + x^3 + \cdots + x^n + \ldots$	$(-1,\ 1)$
12	$\dfrac{1}{1 + x}$	$1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \ldots$	$(-1,\ 1)$

№	Funksiya $f(x)$	Makloren qatori yoyilmasi ($x_0 = 0$)	Yaqinlashish sohasi
13	$\sqrt{1+x}$	$1 + \frac{1}{2}x - \frac{1}{2 \cdot 4}x^2 + \frac{1 \cdot 3}{2 \cdot 4 \cdot 6}x^3 - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 8}x^4 + \dots$	$[-1,\ 1]$
14	$\frac{1}{\sqrt{1+x}}$	$1 - \frac{1}{2}x + \frac{1 \cdot 3}{2 \cdot 4}x^2 - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}x^3 + \dots$	$(-1,\ 1]$

XI BOB

Karrali va Sirt Integrallarining Fizik va Geometrik Tatbiqlari

Manba: `centr_tjazhesti_ploskoi_figury.html` va `poverhnostnye_integraly.html`.

Mavzu 26. Tekis Shakl va Jismning Og'irlik Markazi, Massasi va Inersiya Momentlari

Tekislikdagi D sohani egallagan plastinkaning sirt zichligi $\mu(x,y)$ bo'lsin (agar plastinka **bir jinsli** bo'lsa, $\mu = \text{const} = 1$ deb olinadi va massa m o'rniga yuza S ishlatiladi):

Fizik-geometrik kattalik	Ikki karrali integral (Tekis plastinka D)	Bir jinsli shakl uchun ($\mu = \text{const}$)
Massa m (Yuza S)	$m = \iint_D \mu(x,y)\,dx\,dy$	$S = \iint_D dx\,dy$
Statik momentlar (M_x, M_y)	$M_x = \iint_D y\,\mu(x,y)\,dx\,dy, \quad M_y = \iint_D x\,\mu(x,y)\,dx\,dy$	$M_x = \iint_D y\,dx\,dy, \quad M_y = \iint_D x\,dx\,dy$
Og'irlik markazi (x_C, y_C)	$x_C = \dfrac{M_y}{m}, \qquad y_C = \dfrac{M_x}{m}$	$x_C = \dfrac{1}{S}\iint_D x\,dx\,dy, \quad y_C = \dfrac{1}{S}\iint_D y\,dx\,dy$
Inersiya momentlari (I_x, I_y, I_0)	$I_x = \iint_D y^2\mu(x,y)dxdy, \quad I_y = \iint_D x^2\mu(x,y)dxdy$	$I_0 = I_x + I_y = \iint_D (x^2+y^2)\mu(x,y)dxdy$

Fazoviy jism V uchun ($\rho(x,y,z)$ zichlik bilan): $m = \iiint_V \rho\,dxdydz$, $x_C = \frac{1}{m}\iiint_V x\rho\,dxdydz$, $y_C = \frac{1}{m}\iiint_V y\rho\,dxdydz$, $z_C = \frac{1}{m}\iiint_V z\rho\,dxdydz$.

📝 **Namuna misol (Mathprofi.ru — centr_tjazhesti_ploskoi_figury.html)**

Masala: $y = 4 - x^2$ parabola va $y = 0$ (Ox o'qi) bilan chegaralangan bir jinsli tekis shaklning og'irlik markazi (x_C, y_C) ni toping.

Yechish: 1) Shakl Oy o'qiga nisbatan simmetrik bo'lgani uchun **darhol** $x_C = 0$**!**

2) Shakl yuzi: $S = \int_{-2}^{2}(4-x^2)dx = 2\int_0^2(4-x^2)dx = 2\left[4x - \frac{x^3}{3}\right]_0^2 = \dfrac{32}{3}$.

3) M_x statik moment: $M_x = \iint_D y\,dx\,dy = \int_{-2}^{2}dx\int_0^{4-x^2} y\,dy = \frac{1}{2}\int_{-2}^{2}(4-x^2)^2dx = \int_0^2(16-8x^2+x^4)dx = \left[16x - \frac{8}{3}x^3 + \frac{x^5}{5}\right]_0^2 = \dfrac{256}{15}$.

4) $y_C = \dfrac{M_x}{S} = \dfrac{256/15}{32/3} = \dfrac{8}{5} = 1.6$. Demak, og'irlik markazi: $C(0;\ 1.6)$.

Mavzu 27. Birinchi va ikkinchi tur sirt integrallari (`poverhnostnye_integraly.html`)

Agar Σ sirt $z = z(x,y)$ tenglama bilan berilgan bo'lsa va uning Oxy tekislikdagi proyeksiyasi D soha bo'lsa, sirt yuzasi differensiali (elementi):

$$dS = \sqrt{1 + (z'_x)^2 + (z'_y)^2}\,dx\,dy$$

1-tur sirt integrali (sirt yuzasi bo'yicha):

$$\iint_\Sigma f(x,y,z)\,dS = \iint_D f\big(x,y,z(x,y)\big)\sqrt{1+(z'_x)^2+(z'_y)^2}\,dx\,dy$$

Xususan, $f(x,y,z) = 1$ bo'lganda Σ sirtning yuzi $S(\Sigma) = \iint_\Sigma dS$ kelib chiqadi.

XII BOB

Kompleks Analiz (TFKP), Konform Akslantirishlar va Laplas Jadvali

Manba: `oblasti_na_kompleksnoy_ploskosti.html`, `linii_na_kompleksnoy_ploskosti.html`, `konformnoe_otobrazhenie.html` va `tablica_originalov_j_izobrazhenij.pdf`.

Mavzu 28. Kompleks tekislikda chiziqlar va sohalar (`oblasti_na_kompleksnoy_ploskosti.html`)

Kompleks tenglama / tengsizlik	Geometrik ma'nosi ($z = x + iy,\ z_0 = x_0 + iy_0$)
$\lvert z - z_0\rvert = R$	Markazi z_0 nuqtada, radiusi R bo'lgan **aylana**: $(x-x_0)^2 + (y-y_0)^2 = R^2$
$\lvert z - z_0\rvert < R$ (yoki $\le R$)	Markazi z_0, radiusi R bo'lgan **ochiq (yoki yopiq) doira**
$r < \lvert z - z_0\rvert < R$	Markazi z_0 da bo'lgan **halqa (ikki bog'lamli soha)**
$\lvert z - z_1\rvert = \lvert z - z_2\rvert$	z_1 va z_2 nuqtalarni tutashtiruvchi kesmaning **o'rta perpendikulyari** (to'g'ri chiziq)
$\lvert z - F_1\rvert + \lvert z - F_2\rvert = 2a$ $(2a > \lvert F_1F_2\rvert)$	Fokuslari F_1 va F_2 nuqtalarda, katta o'qi $2a$ bo'lgan **ellips**
$\big\lvert \lvert z - F_1\rvert - \lvert z - F_2\rvert \big\rvert = 2a$ $(2a < \lvert F_1F_2\rvert)$	Fokuslari F_1 va F_2 nuqtalarda bo'lgan **giperbola**
$\alpha < \arg(z - z_0) < \beta$	Uchi z_0 nuqtada bo'lgan, α va β nurlar orasidagi **burchak sohasi**
$a \le \operatorname{Re}(z) \le b$ (yoki $c \le \operatorname{Im}(z) \le d$)	$x = a$ va $x = b$ (yoki $y = c$ va $y = d$) orasidagi **vertikal (yoki gorizontal) polosa**

Mavzu 29. Konform akslantirishlar: Cho'zilish koeffitsiyenti, burilish burchagi va obraz yuzi (`konformnoe_otobrazhenie.html`)

Agar $w = f(z)$ funksiya D sohada analitik bo'lsa va $f'(z_0) \neq 0$ bo'lsa, u z_0 nuqtada **konform akslantirish** (burchaklarni va cheksiz kichik shakllar o'xshashligini saqlovchi akslantirish) amalga oshiradi:

- **Chiziqli cho'zilish (masshtab) koeffitsiyenti:** $k = \lvert f'(z_0)\rvert$ ($k > 1$ — cho'zilish, $k < 1$ — siqilish).
- **Burilish burchagi:** $\alpha = \arg f'(z_0)$ ($\alpha > 0$ — soat miliga teskari, $\alpha < 0$ — soat mili bo'ylab).
- **L chiziq obrazining uzunligi $\mathcal{L}$ va D soha obrazining yuzi $\mathcal{S}$:**

$$\mathcal{L} = \int_L |f'(z)|\,|dz|, \qquad \mathcal{S} = \iint_D |f'(z)|^2\,dx\,dy$$

Namuna misol (Mathprofi.ru — konformnoe_otobrazhenie.html)

Masala: $w = z^2$ akslantirishda $z_0 = 1 + i$ nuqtadagi cho'zilish koeffitsiyenti va burilish burchagini toping.

Yechish: $f'(z) = 2z \implies f'(1+i) = 2 + 2i$. U holda:

1) Cho'zilish koeffitsiyenti: $k = |2 + 2i| = \sqrt{2^2 + 2^2} = 2\sqrt{2} \approx 2.83$ marta cho'ziladi.

2) Burilish burchagi: $\alpha = \arg(2 + 2i) = \arctan\dfrac{2}{2} = \dfrac{\pi}{4}$ (45° soat miliga teskari buriladi).

Mavzu 30. Laplas Almashtirishi: Originallar va Tasvirlarning To'liq Jadvali (`tablica_originalov_j_izobrazhenij.pdf`)

№	Original $f(t)$ $(t \ge 0)$	Tasvir $F(p) = \int_0^\infty e^{-pt} f(t)dt$	№	Original $f(t)$ $(t \ge 0)$	Tasvir $F(p)$
1	1 (Xevisayd $\eta(t)$)	$\dfrac{1}{p}$	10	$e^{\alpha t}\sin\omega t$	$\dfrac{\omega}{(p-\alpha)^2 + \omega^2}$
2	t	$\dfrac{1}{p^2}$	11	$e^{\alpha t}\cos\omega t$	$\dfrac{p-\alpha}{(p-\alpha)^2 + \omega^2}$
3	t^n $(n \in \mathbb{N})$	$\dfrac{n!}{p^{n+1}}$	12	$t\sin\omega t$	$\dfrac{2\omega p}{(p^2 + \omega^2)^2}$

№	Original $f(t)$ $(t \geq 0)$	Tasvir $F(p) = \int_0^\infty e^{-pt} f(t) dt$	№	Original $f(t)$ $(t \geq 0)$	Tasvir $F(p)$
4	$e^{\alpha t}$	$\dfrac{1}{p-\alpha}$	13	$t\cos\omega t$	$\dfrac{p^2-\omega^2}{(p^2+\omega^2)^2}$
5	$t^n e^{\alpha t}$	$\dfrac{n!}{(p-\alpha)^{n+1}}$	14	$\sin\omega t - \omega t\cos\omega t$	$\dfrac{2\omega^3}{(p^2+\omega^2)^2}$
6	$\sin(\omega t)$	$\dfrac{\omega}{p^2+\omega^2}$	15	$\sin\omega t + \omega t\cos\omega t$	$\dfrac{2\omega p^2}{(p^2+\omega^2)^2}$
7	$\cos(\omega t)$	$\dfrac{p}{p^2+\omega^2}$	16	$f'(t)$ (1-hosila)	$pF(p) - f(0)$
8	$\sinh(\omega t)$	$\dfrac{\omega}{p^2-\omega^2}$	17	$f''(t)$ (2-hosila)	$p^2F(p) - pf(0) - f'(0)$
9	$\cosh(\omega t)$	$\dfrac{p}{p^2-\omega^2}$	18	$f^{(n)}(t)$ (n-hosila)	$p^nF(p) - p^{n-1}f(0) - \cdots - f^{(n-1)}(0)$

XIII BOB

Ehtimollar Nazariyasi va Matematik Statistikaning Maxsus Mavzulari hamda Jadvallari

Manba: `formuly_kombinatoriki.pdf`, `formuly_teorii_veroyatnostey.pdf`, `asimmetriya_i_excess.html`, `ocenki_po_povtornoy_i_bespovtornoy_vyborke.html`, `gruppirovka_dannyh.html`, `koefficient_korrelyacii_fehnera.html`, `uravnenie_mnozhestvennoy_regressii.html` va `ryady_dinamiki.html`.

Mavzu 31. Kombinatorika (Takrorlanuvchi va Takrormas) hamda Taqsimotlar Yig'ma Jadvali

Kombinatorik birikma turi	Takrorlanmasdan (Oddiy)	Takrorlanish bilan (С повторениями)
O'rin almashtirish (Перестановки) (Tartib muhim, barcha n ta element qatnashadi)	$P_n = n! = 1 \cdot 2 \cdot 3 \cdots n$ $(0! = 1)$	$P_n(n_1, n_2, \ldots, n_k) = \dfrac{n!}{n_1! \cdot n_2! \cdots n_k!}$ $(n_1 + n_2 + \cdots + n_k = n)$
O'rinlashtirish (Размещения) (n tadan m ta tanlanadi, *tartib muhim*)	$A_n^m = \dfrac{n!}{(n-m)!} = n(n-1)\cdots(n-m+1)$	$\bar{A}_n^m = n^m$
Guruhlash / Kombinatsiya (Сочетания) (n tadan m ta tanlanadi, *tartib muhim emas*)	$C_n^m = \dfrac{n!}{m!(n-m)!}, \quad C_n^m = C_n^{n-m}$	$\bar{C}_n^m = C_{n+m-1}^m = \dfrac{(n+m-1)!}{m!(n-1)!}$

Asosiy Diskret va Uzluksiz Taqsimotlarning Xarakteristikalari (`formuly_teorii_veroyatnostey.pdf`)

Taqsimot nomi	Ehtimollik qonuni / Zichlik $f(x)$	Matematik kutilma $M(X)$	Dispersiya $D(X)$
Binomial $B(n,p)$	$P_n(k) = C_n^k p^k q^{n-k}$ $(k = 0, \ldots, n)$	$M(X) = np$	$D(X) = npq$
Puasson $\mathrm{Pois}(\lambda)$	$P(X = k) = \dfrac{\lambda^k e^{-\lambda}}{k!}$ $(\lambda = np)$	$M(X) = \lambda$	$D(X) = \lambda$
Geometrik $G(p)$	$P(X = k) = q^{k-1}p$ $(k = 1, 2, \ldots)$	$M(X) = \dfrac{1}{p}$	$D(X) = \dfrac{q}{p^2}$
Gipergeometrik	$P(X = m) = \dfrac{C_M^m \cdot C_{N-M}^{n-m}}{C_N^n}$	$M(X) = n\dfrac{M}{N}$	$D(X) = n\dfrac{M}{N}\left(1 - \dfrac{M}{N}\right)\dfrac{N-n}{N-1}$
Tekis (Uzluksiz) $U(a,b)$	$f(x) = \dfrac{1}{b-a}$, $x \in [a,b]$	$M(X) = \dfrac{a+b}{2}$	$D(X) = \dfrac{(b-a)^2}{12}$
Ko'rsatkichli $\mathrm{Exp}(\lambda)$	$f(x) = \lambda e^{-\lambda x}$, $x \geq 0$	$M(X) = \dfrac{1}{\lambda}$	$D(X) = \dfrac{1}{\lambda^2}, \quad \sigma(X) = \dfrac{1}{\lambda}$
Normal (Gauss) $N(a, \sigma^2)$	$f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-a)^2}{2\sigma^2}}$	$M(X) = a$	$D(X) = \sigma^2$

Mavzu 32. Empirik Taqsimotning Asimmetriya (A_s) va Ekstsess (E_k) Koeffitsiyentlari (`asimmetriya_i_excess.html`)

Variatsion qatorning normal taqsimotdan qanchalik og'ganligini baholash uchun 3- va 4-tartibli empirik markaziy momentlar μ_3, μ_4 hisoblanadi:

$$\mu_3 = \frac{1}{n}\sum (x_i - \bar{x}_v)^3 n_i, \qquad \mu_4 = \frac{1}{n}\sum (x_i - \bar{x}_v)^4 n_i$$

- **Asimmetriya koeffitsiyenti (Qiyshiqlik ko'rsatkichi):**

$$A_s = \frac{\mu_3}{\sigma_v^3}$$

Agar $A_s = 0$ — simmetrik; $A_s > 0$ — **o'ng tomonlama asimmetriya** (uzun «dum» o'ngda, $\bar{x}_v > Me > Mo$); $A_s < 0$ — **chap tomonlama asimmetriya** (uzun «dum» chapda). $|A_s| < 0.25$ — kuchsiz, $0.25 \leq |A_s| \leq 0.5$ — o'rtacha, $|A_s| > 0.5$ — kuchli asimmetriya.

- **Ekstsess koeffitsiyenti (Cho'qqidorlik ko'rsatkichi):**

$$E_k = \frac{\mu_4}{\sigma_v^4} - 3$$

Normal taqsimot uchun $E_k = 0$. Agar $E_k > 0$ bo'lsa, taqsimot grafigi normal egri chiziqdan **o'tkirroq (balandroq) cho'qqili**; $E_k < 0$ bo'lsa, **yalpoq (past) cho'qqili** bo'ladi.

Mavzu 33. Takroriy va Takrormas Tanlanmalarda Bosh O'rtacha va Bosh Ulushni Baholash (`ocenki_po_povtornoy_i_bespovtornoy_vyborke.html`)

Hajmi N bo'lgan bosh to'plamdan hajmi n bo'lgan tanlanma olinganda ishonchlilik intervali $(\bar{x}_v - \Delta;\ \bar{x}_v + \Delta)$ va $(w - \Delta_p;\ w + \Delta_p)$ dagi chekli xatolik $\Delta = t_\gamma \cdot \mu$ tanlanma turiga qarab farq qiladi:

Ko'rsatkich	Takroriy (Qaytariluvchi) tanlanma	Takrormas (Qaytarilmaydigan) tanlanma
Bosh o'rtacha $\bar{x}_0$ uchun chekli xatolik $\Delta_{\bar{x}}$	$\Delta_{\bar{x}} = t\sqrt{\frac{\sigma^2}{n}}$	$\Delta_{\bar{x}} = t\sqrt{\frac{\sigma^2}{n}\left(1 - \frac{n}{N}\right)}$
Bosh ulush p (Binomial ehtimol) uchun Δ_p ($w = m/n$ — nisbiy chastota)	$\Delta_p = t\sqrt{\frac{w(1-w)}{n}}$	$\Delta_p = t\sqrt{\frac{w(1-w)}{n}\left(1 - \frac{n}{N}\right)}$
Berilgan Δ aniqlik uchun zaruriy tanlanma hajmi n	$n_{\bar{x}} = \frac{t^2\sigma^2}{\Delta^2}, \qquad n_p = \frac{t^2 w(1-w)}{\Delta_p^2}$	$n_{\bar{x}} = \frac{Nt^2\sigma^2}{N\Delta^2 + t^2\sigma^2}, \qquad n_p = \frac{Nt^2 w(1-w)}{N\Delta_p^2 + t^2 w(1-w)}$

Mavzu 34. Statistik Ma'lumotlarni Guruhlash va Qayta Guruhlash (`gruppirovka_dannyh.html`, `analiticheskaya_gruppirovka.html`)

- **Tipologik guruhlash:** Bir jinsli bo'lmagan to'plamni sifat jihatidan bir jinsli guruhlarga ajratish (masalan, korxonalarni mulkchilik shakli bo'yicha).
- **Strukturaviy guruhlash:** Bir jinsli to'plamning ichki tuzilishini o'rganish (variatsion qatorlar).
- **Analitik va Kombinatsion guruhlash:** Omil (X) va natijaviy (Y) belgilar o'rtasidagi bog'liqlikni guruh o'rtachalari $\bar{y}_j$ orqali aniqlash (2 va undan ortiq omil bo'yicha guruhlash — *kombinatsion guruhlash*).
- **Ikkilamchi qayta guruhlash (Перегруппировка):** Turli intervalli qatorlarni taqqoslash uchun ularni foiz ulushlari bo'yicha yoki yangi yagona intervallarga proporsional bo'lish orqali qayta guruhlash.

Mavzu 35. Fexner Ishoralar Korrelyatsiyasi Koeffitsiyenti (`koefficient_korrelyacii_fehnera.html`)

X va Y belgilar o'rtasidagi bog'liqlik yo'nalishini va kuchini eng tez baholovchi **noparametrik ko'rsatkich — Fexner koeffitsiyenti (K_Φ)**:

1. $\bar{x}$ va $\bar{y}$ o'rtachalar hisoblanadi.
2. Har bir x_i uchun $x_i - \bar{x}$ ning ishorasi ($+$ yoki $-$) va har bir y_i uchun $y_i - \bar{y}$ ning ishorasi yoziladi.
3. Ishoralar bir xil ($++$ yoki $--$) bo'lgan juftliklar soni C (совпадения) va ishoralar har xil ($+-$ yoki $-+$) bo'lgan juftliklar soni H (несовпадения) sanaladi ($C + H = n$).
4. **Fexner formulasi:**

$$K_\Phi = \frac{C - H}{C + H} = \frac{C - H}{n}, \qquad -1 \le K_\Phi \le 1$$

Namuna misol (Mathprofi.ru): Fexner koeffitsiyentini topish

Masala: $X = (2, 4, 5, 7, 12) \implies \bar{x} = 6$ va $Y = (10, 14, 18, 22, 26) \implies \bar{y} = 18$ uchun K_Φ ni toping.

Yechish: $\operatorname{sign}(x_i - 6) = (-, -, -, +, +)$, $\operatorname{sign}(y_i - 18) = (-, -, +, +, +)$. Mos kelishlar: 1-, 2-, 4-, 5-juftliklar ($C = 4$), mos kelmaslik: 3-juftlik ($H = 1$). Demak, $K_\Phi = \frac{4 - 1}{4 + 1} = \frac{3}{5} = +0.6$ (o'rtacha to'g'ri bog'lanish).

Mavzu 36. Ko'p Omilli Chiziqli Regressiya Modeli (`uravnenie_mnozhestvennoy_regressii.html`)

Ikki omilli chiziqli regressiya tenglamasi: $\hat{y} = b_0 + b_1 x_1 + b_2 x_2$. Juft korrelyatsiya koeffitsiyentlari $r_{yx_1}, r_{yx_2}, r_{x_1x_2}$ va standart chetlanishlar $\sigma_y, \sigma_{x_1}, \sigma_{x_2}$ ma'lum bo'lganda barcha parametrlar quyidagi formulalar bilan topiladi:

- **Standartlashtirilgan β-koeffitsiyentlar va regressiya koeffitsiyentlari b_1, b_2:**

$$\beta_1 = \frac{r_{yx_1} - r_{yx_2} r_{x_1x_2}}{1 - r_{x_1x_2}^2}, \quad \beta_2 = \frac{r_{yx_2} - r_{yx_1} r_{x_1x_2}}{1 - r_{x_1x_2}^2}, \quad b_1 = \beta_1 \frac{\sigma_y}{\sigma_{x_1}}, \quad b_2 = \beta_2 \frac{\sigma_y}{\sigma_{x_2}}, \quad b_0 = \bar{y} - b_1 \bar{x}_1 - b_2 \bar{x}_2$$

- **Elastiklik koeffitsiyentlari (E_i — x_i omil 1 ga o'zgarganda y necha foizga o'zgarishi):**

$$E_1 = b_1 \frac{\bar{x}_1}{\bar{y}}, \qquad E_2 = b_2 \frac{\bar{x}_2}{\bar{y}}$$

- **Xususiy korrelyatsiya koeffitsiyenti (x_2 ta'siri chiqarib tashlanganda y va x_1 bog'liqligi):**

$$r_{yx_1 \cdot x_2} = \frac{r_{yx_1} - r_{yx_2} \cdot r_{x_1x_2}}{\sqrt{(1 - r_{yx_2}^2)(1 - r_{x_1x_2}^2)}}$$

- **Ko'p omilli determinatsiya koeffitsiyenti (R^2) va Fisher F-mezoni:**

$$R^2_{yx_1x_2} = \beta_1 r_{yx_1} + \beta_2 r_{yx_2} = \frac{r_{yx_1}^2 + r_{yx_2}^2 - 2 r_{yx_1} r_{yx_2} r_{x_1x_2}}{1 - r_{x_1x_2}^2}, \qquad F_{\text{kuz}} = \frac{R^2}{1 - R^2} \cdot \frac{n - m - 1}{m}$$

Mavzu 37. Dinamika Qatorlarining Bazisli, Zanjirli va O'rtacha Ko'rsatkichlari (`ryady_dinamiki.html`)

$y_1, y_2, \ldots, y_n$ vaqt qatori uchun bazisli (boshlang'ich y_1 ga nisbatan) va zanjirli (oldingi y_{i-1} ga nisbatan) ko'rsatkichlar:

Ko'rsatkich nomi	Bazisli usul	Zanjirli usul	O'rtacha ko'rsatkich ($i = 1, \ldots, n$)
Mutlaq o'sish (Δ)	$\Delta_i^b = y_i - y_1$	$\Delta_i^z = y_i - y_{i-1}$	$\bar{\Delta} = \dfrac{y_n - y_1}{n-1} = \dfrac{\sum \Delta_i^z}{n-1}$
O'sish koeffitsiyenti ($K_{\text{o'sh}}$)	$K_i^b = \dfrac{y_i}{y_1}$	$K_i^z = \dfrac{y_i}{y_{i-1}}$	$\bar{K}_{\text{o'sh}} = \sqrt[n-1]{\dfrac{y_n}{y_1}} = \sqrt[n-1]{K_2^z \cdot K_3^z \cdots K_n^z}$
O'sish sur'ati ($T_{\text{o'sh}}$,)	$T_i^b = \dfrac{y_i}{y_1} \cdot 100$	$T_i^z = \dfrac{y_i}{y_{i-1}} \cdot 100$	$\bar{T}_{\text{o'sh}} = \bar{K}_{\text{o'sh}} \cdot 100$
Qo'shimcha o'sish sur'ati ($T_{\text{qo'sh}}$,)	$T_{\text{qo'sh},i}^b = T_i^b - 100$	$T_{\text{qo'sh},i}^z = T_i^z - 100$	$\bar{T}_{\text{qo'sh}} = \bar{T}_{\text{o'sh}} - 100$
1 **qo'shimcha o'sishning mutlaq qiymati (A_i)**	—	$A_i = \dfrac{\Delta_i^z}{T_{\text{qo'sh},i}^z} = 0.01 \cdot y_{i-1}$	—